Harshit Chhabi

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Education

Vellore Institute of Technology, Vellore 2024 – 2028
B.Tech in Computer Science and Engineering (Information Security)
CGPA: 9.00 / 10 Semester GPAs: 9.12, 9.31, 8.94, 8.73 Class XII CBSE: 89.8% Class X CBSE: 94.8%

Technical Skills

Languages: Go, TypeScript, JavaScript, Python, C, C++, Java, SQL
Frameworks: Next.js, React, Node.js, Express.js, Fastify, FastAPI, Tailwind, WebSockets/Socket.io, Drizzle ORM
Cloud & DevOps: AWS (EC2, ALB, Auto Scaling, RDS, ElastiCache, IAM), Docker, Docker Compose, Caddy, Linux
Tools: PostgreSQL, MySQL, Redis, scikit-learn, PyTorch, OpenCV, MediaPipe, bcrypt, Jest, Git/GitHub, VS Code
Security: Web/App Security, VAPT, Threat Analysis, Cryptography (AES, RSA, SHA), Secure Coding, Access Control

Experience

Cybersecurity Analyst Intern May 2025 – July 2025
State Bank of India, Global IT Centre

- Performed web and application security testing, access control validation and threat analysis (VAPT).
- Applied encryption standards and secure coding practices in a regulated banking environment.
- Built the Secure File Hashing & Dual Encryption web app as the internship project.

Technical Head 2026 – Present
Dream Merchants Club, VIT Vellore – built the platforms for the club's events

- **Stockastic**: live stock-market simulation; rewrote backend from TypeScript to Go; Next.js, AWS.
- **Dream Team**: real-time IPL-style player auction; Go, Next.js, PostgreSQL, WebSockets, Docker.
- **Bricks by Bid**: auction and city-building game; Next.js, Node.js, PostgreSQL, Drizzle ORM.

Projects

AI-Powered Network Intrusion Detection & Deception System github.com/harshitchhabi/IDS

- ML IDS (random forest, XGBoost) on CICIDS2017 detecting DDoS, port-scan and brute-force traffic.
- Cowrie SSH honeypot with a real-time alert dashboard (Python, FastAPI, HTMX, Chart.js).
- Showed honeypot auto-labeling enables data poisoning; defense ∼781× cheaper than kNN.
- Findings generalized to UNSW-NB15 and two more models (logistic regression, MLP).

Signify: Sign Language to Text & Speech github.com/harshitchhabi/Signify

- Real-time translation from live video using MediaPipe, OpenCV and TTS APIs.
- PyTorch / Scikit-learn gesture classifier; FastAPI backend, Next.js frontend.
- Sliding-window majority voting for stable predictions; supports video upload.

Secure File Hashing & Dual Encryption github.com/harshitchhabi/File-hash

- SHA-256 hashing with AES-256 + RSA-2048 hybrid encryption and tamper detection.
- Node.js, Express, MySQL, bcrypt session auth.
- Dashboard to upload, download, decrypt and delete files, with MySQL file history.

Vayu Rakshak: HCHO Pollution Hotspot Detection github.com/harshitchhabi/S5P

- Detects HCHO hotspots from Sentinel-5P TROPOMI data to track Punjab/Haryana stubble burning.
- Python, scikit-learn, XGBoost, Streamlit.
- Cross-checks hotspots against NASA FIRMS fire data and correlates them with AQI.

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle, 2025
- SQL (Intermediate) – HackerRank, 2026

Achievements / Hackathons

- Bharatiya Antariksh Hackathon 2026, Team GEMs (participant) – Vayu Rakshak, now being extended for SIH 2026